

DS 2002: EV Charging Station Analytics and Cloud Pipelines

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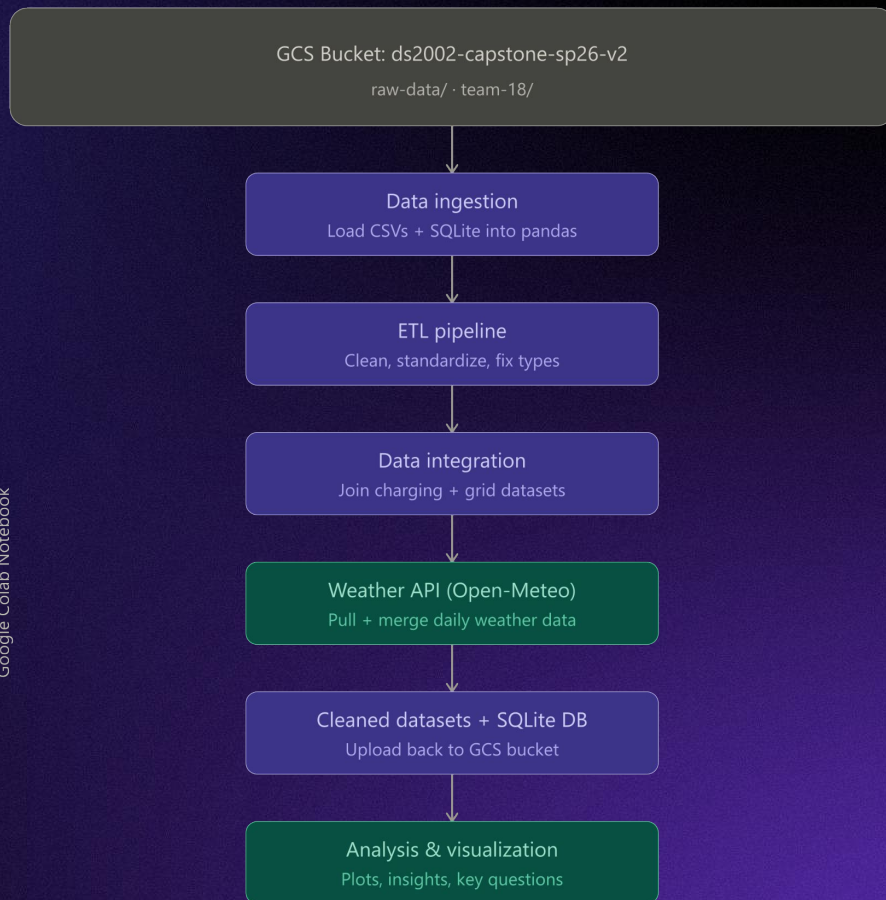
Recommendation

Cloud Pipeline Overview

Cloud Pipeline Overview

- Pull data from GCP bucket
- Run our ETL pipeline
 - Important cleaning decisions
- Ingest our data for analysis
- Pull additional data from Open-Meteo
- Combine everything and perform analysis
- Upload our cleaned datasets back to bucket (visualization is out of order)

Google Colab Notebook



Data Cleaning Decisions

Vehicle ID Mapping

- One important data cleaning decisions we had to make was how to standardize vehicle IDs
- Resolved this by constructing a Python dictionary mapping each of the differently-formatted vehicle IDs that referenced the same vehicle to one singular vehicle ID format
 - VH-XXX
- Applied the mapping to each dataframe to standardize!

Before

vehicle_id
VH-008
VH-005
VEH#0005
VH-005
VH-011
...
V_bmw_ix
VH-001
VH-010
VH-004
VH-004

After

vehicle_id
VH-008
VH-005
VH-005
VH-005
VH-011
...
VH-011
VH-001
VH-010
VH-004
VH-004

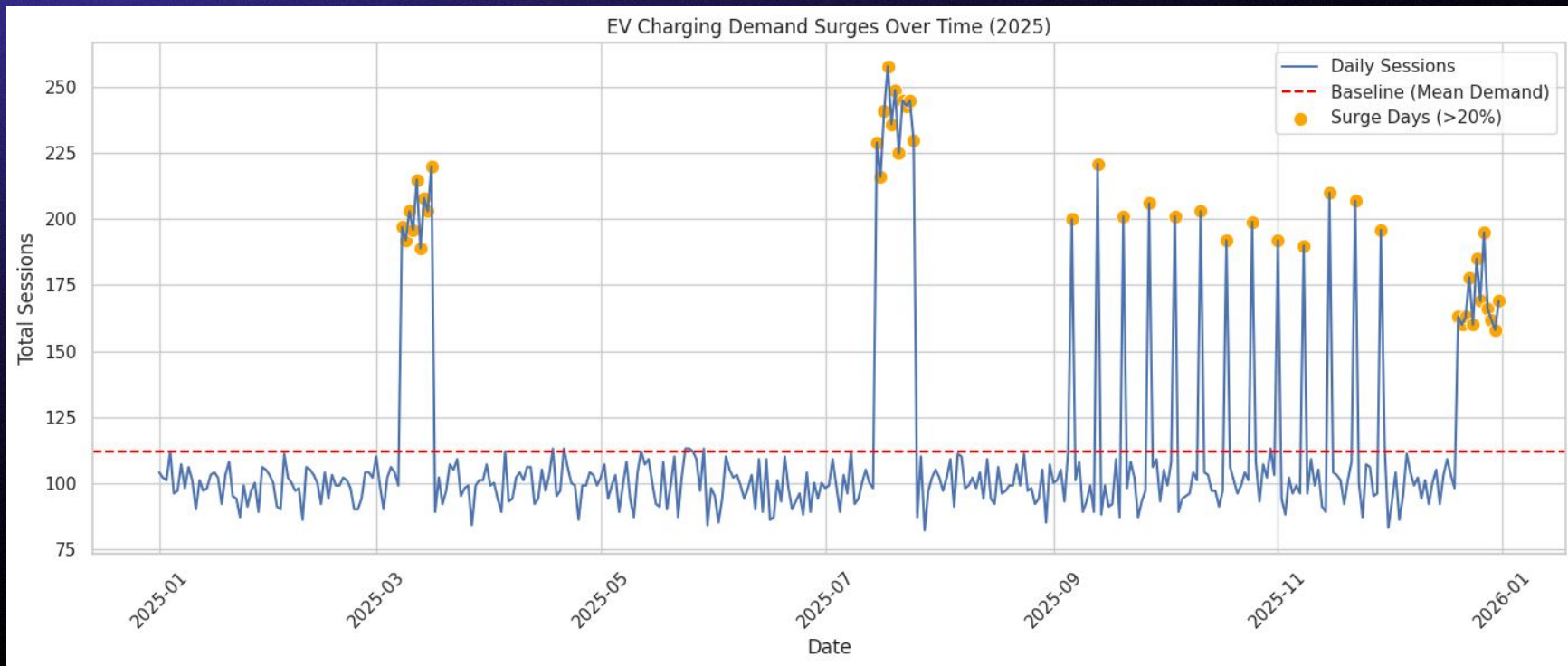
Station ID Mapping

- Similar process as vehicles, but for stations
- Multiple station IDs did often point to the same station, but were also malformed
- Resolved using a similar process as vehicles
 - Map each malformed station to its standardized format
 - STN-XXX
- Acceptable change because there are a limited number of stations

```
station_mapping = {  
    "STN001": "STN-001",  
    "STN01": "STN-001",  
    "STN002": "STN-002",  
    "STN02": "STN-002",  
    "STN003": "STN-003",  
    "STN03": "STN-003",  
    "STN004": "STN-004",  
    "STN04": "STN-004",  
    "STN005": "STN-005",  
    "STN05": "STN-005",  
    "STN006": "STN-006",  
    "STN06": "STN-006",  
    "STN007": "STN-007",  
    "STN07": "STN-007",  
    "STN_007": "STN-007",  
    "STN008": "STN-008",  
    "STN08": "STN-008",  
    "STN009": "STN-009",  
    "STN09": "STN-009",  
    "STN010": "STN-010",  
    "STN10": "STN-010"  
}
```

Visualization Analysis

Question 1: Demand Surge Identification

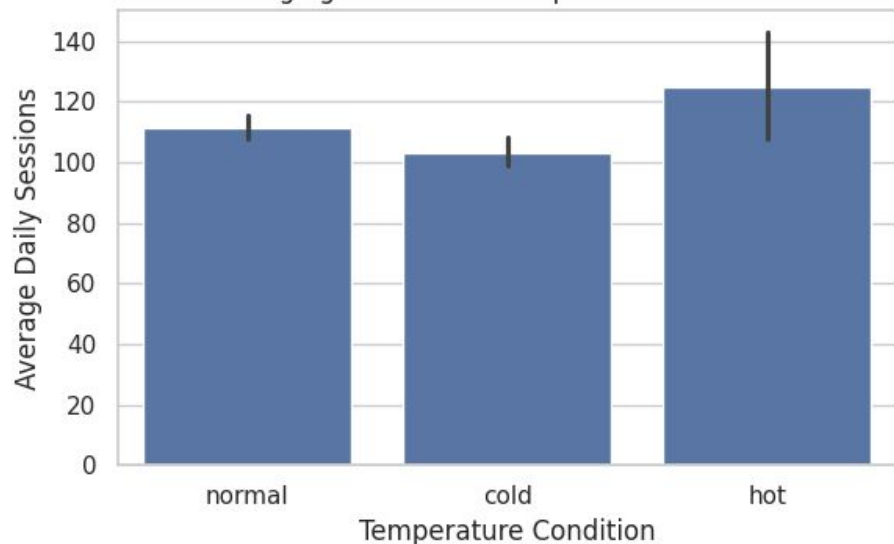


Question 1: Demand Surge Identification

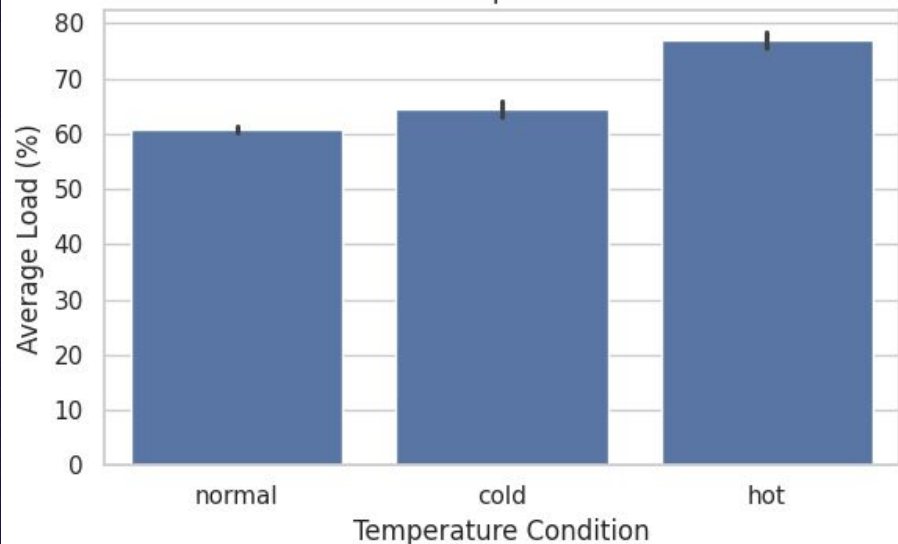
- Measured demand by number of charging sessions per day
 - Did NOT use kwh consumed to avoid abnormal sessions from tainting data
- Baseline surge is average number of sessions per day for the year
- Observations
 - Surge in demand in March
 - Peak in mid-July
 - Consistent surge days after September through the rest of the year

Question 3: Weather and Grid Correlation

Charging Demand vs Temperature Extremes



Grid Load vs Temperature Extremes



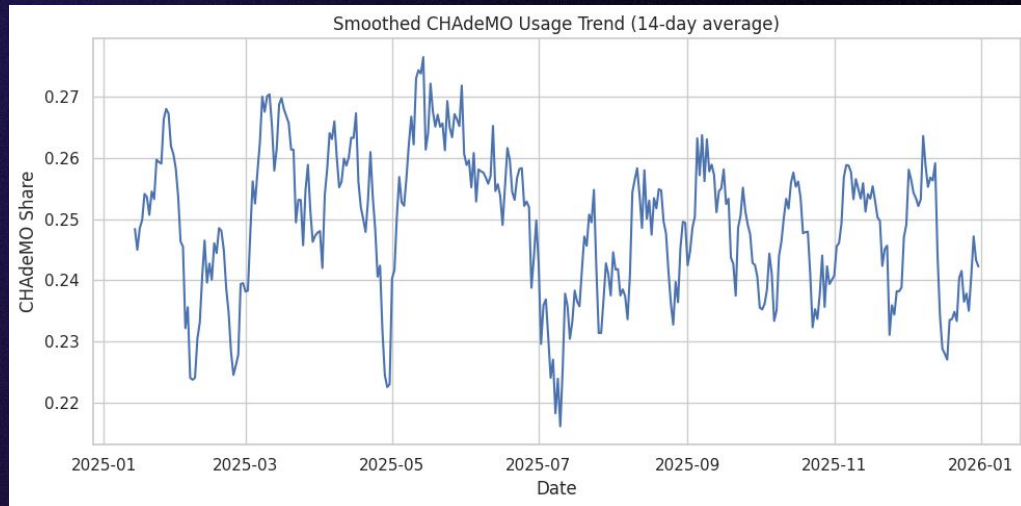
Question 3: Weather and Grid Correlation

- Integrated API temperature data with charging demand
- Calculated daily temperatures outside of 0.10 and 0.90 quantiles as “extreme”
 - Further broken down into “cold”, “hot”, “normal”
- No observable correlation between extremes and charging demand
- Similar classification process for grid load, except we use integrated temperature data
- Both “cold” and “hot” days correspond with higher grid load than “normal” days
 - Correlation!

CHAdemo Recommendation

Question 5: The Connector Type Investigation

- Assign each session a flag of whether it uses CHAdeMO or not
- Compute the proportion of daily sessions using CHAdeMO
- Across the year, **no observable steady decline in usage**
 - For both raw numbers and rolling averages
- The city should invest because it shows no direct signs of decreased usage!



Thank you!